Table of Contents

1. CNC lathe

1.1. Scope and Purpose:

1.1.1. The purpose of this Standard Operating Procedure (SOP) is to provide authorized students with clear instructions for safely operating the Haas TL-1 CNC lathe in an educational laboratory. The TL-1 is used to demonstrate industrial CNC turning principles, machine controls, work holding, tooling, and safe manufacturing practices.

1.1.2. This SOP enables a student with limited CNC experience to run a specific demonstration program that has been reviewed and approved by a qualified instructor or laboratory supervisor. It is intended to promote consistent operation, reduce the risk of injury or equipment damage, and identify the conditions under which operation must be stopped.

1.1.3. This SOP supplements, but does not replace, hands-on instruction, laboratory safety requirements, machine safety decals, or the Haas Lathe Operator's Manual.

1.1.4. This SOP applies to authorized students, instructors, laboratory supervisors, and other personnel operating the Haas TL-1 CNC lathe in the Makino Subtractive lab.

1.1.5. It covers:

- Pre-operation safety requirements and machine inspection.
- Machine startup and zero-return procedures.
- Loading a pre-approved workpiece.
- Selecting and verifying an instructor-approved demonstration program.
- Conducting a supervised first run.
- Running the approved program.
- Removing the completed workpiece.
- Cleaning the machining area.
- Safely shutting down the machine.
- Responding to alarms, unexpected motion, or other unsafe conditions.

1.1.6. This SOP is limited to running an approved program using a setup that has already been inspected and verified by a qualified instructor or laboratory supervisor. Unless separately trained and authorized, students should not:

- Create or edit CNC programs.
- Establish or change work offsets or tool offsets.
- Install, remove, or adjust cutting tools.
- change chuck jaws or other work holding devices.
- Machine material or part dimensions other than those specified on the approved setup sheet.
- Bypass or alter guards, door switches, interlocks, alarms, or other safety devices.
- Enter the machining area while automatic motion is possible.
- Troubleshoot electrical, hydraulic, pneumatic, mechanical, or control-system faults.
- Clear a collision, jam, or obstruction.
- Perform machine repairs, service, or maintenance.
- Operate the machine without the required supervision.

1.1.7. Any task outside this scope must be performed or directly supervised by a qualified and authorized instructor, machinist, laboratory supervisor, or Haas service representative.

1.2. Authorization and Supervision:

1.2.1. First-time operators must be directly supervised by a qualified TL-1 operator
1.2.2. This SOP permits the operator to run only the specified, pre0approved program and setup
1.2.3. The operator must not edit the program, set or change tool/work offsets, change tools or work holding, bypass safeguards, troubleshoot faults, or recover from a collision or jam unless specifically trained and authorized.
1.2.4. Be aware of your surroundings, and review the operation of the Emergency Stop, Feed Hold, Cycle Start, Reset, and Power Off controls before beginning.

1.3. Safety Warnings, Labels, and Symbols:

The Haas TL-1 contains safety labels that identify hazards, prohibited actions, required precautions, and operating information. Before operating the machine, the operator must locate, read, and understand all safety labels on the machine.

- Never remove, cover, alter, or ignore a safety label. If a label is missing, damaged, unreadable, or not understood, do not operate the machine. Notify the instructor or laboratory supervisor.
- A safety label may combine several symbols to identify the hazard, show an action that is prohibited, and show the correct precaution.

1.3.1. Moving and Rotating Components

Moving parts can entangle, trap, crush, and cut. Keep all parts of your body away from machine parts when they move, or whenever motion is possible. Motion is possible when the power is on and the machine is not in **[EMERGENCY STOP]**. Secure loose clothing, hair, etc. Remember that automatically controlled devices can start at any time.

1.3.2. Chuck and Ejected-Part Hazard

Always securely clamp workpieces in the chuck or collet. Properly fasten chuck jaws.

1.3.3. Unsupported Bar-Stock Hazard

Do not extend unsupported bar stock out the rear of the drawtube. Unsupported bar can bend and “whip”. A whipping bar can cause severe injury or death.

1.3.4. Door and Enclosure Hazard

Do not enter the machine enclosure when the machine is capable of automatic motion. When you must enter the enclosure to complete tasks, press **[EMERGENCY STOP]** or power off the machine. Put a safety tag on the control pendant to alert other people that you are inside the machine, and that they must not turn on or operate the machine

1.3.5. Chip, Dust, Coolant, and Mist Hazard

Machining operations can create hazardous chips, dust or mist. This is function of the materials being cut, the metalworking fluid and cutting tools used and the machining speeds/feeds. It is up to the owner/operator of the machine to determine if personal protective equipment such as safety goggles or a respirator is required and also if a mist extraction system is needed. All enclosed models have a provision for connecting a mist extraction system. Always read and understand the Safety Data Sheets (SDS) for the workpiece material, the cutting tools and the metalworking fluid.

1.3.6. High-Voltage Hazard

There are high voltage components on the machine that can cause electrical shock. Always use care around high voltage components.

1.3.7. Hot-Surface Hazard

The Regen is used by the spindle drive to dissipate excess power and will get hot. Always use care around the Regen

1.3.8. Noise Hazard

Always wear safety glasses or goggles when you are near a machine. Airborne debris can cause eye damage. Always wear hearing protection when you are near a machine. Machine noise can exceed 70 dBA

1.3.9. Entanglement Hazard

Secure loose clothing, hair, jewelry, etc. Do not wear gloves around rotating machine components. You can be pulled into the machine, resulting in severe injury or death. Automatic motion is possible when the power is on and the machine is not in **[EMERGENCY STOP].**

1.3.10. Unrecognized Warning or Symbol

If the operator encounters an unfamiliar alarm, label, warning, or safety symbol:

- Stop the current procedure.
- Do not press Cycle Start or Reset to continue.
- Read the operator’s manual for more detailed information.

- Notify the instructor or laboratory supervisor.
- Resume operation only after the warning has been explained and the machine has been confirmed safe.

Read and understand the operator's manual and other instructions included with your machine.

1.4.Required Personal Protective Equipment (PPE):

- Wear approved safety glasses
- Wear hearing protecting when required by the machining operation or laboratory rules
- Wear close toe shoes
- Secure long hair and remove or secure loose clothing, jewelry, neckties, lanyards, and any other items that could become entangled.
- Do not wear gloves while loading the part, operating the controls, or whenever spindle rotation is possible. Gloves may be worn only when handling sharp chips or complete parts after the spindle and all machine motion has completely stopped

1.5.Machine Startup:

1.1.1.Confirm that the power switch is on (flipped up position) and no tag on the switch (located on the back of the machine)

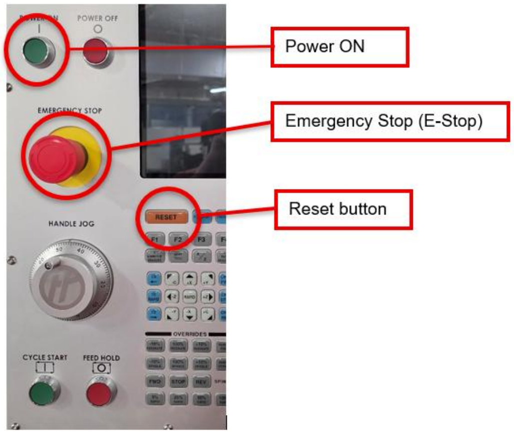
1.1.2.From there look for a blue air valve on the wall near the machine and check if it is open (parallel with the vertical copper pipe). If it is not, then turn the valve from its horizontal position to the vertical position.

1.1.3. After confirming that it's open, check the air pressure gauge next to the valve and back right, bottom of the machine to make sure the indicators are between 80-120PSI as recommended by the manufacture

1.1.4. Confirm that you don't hear any air leaks coming from the machine

1.1.5. Verify that there is nothing on or in the chuck, that there is no tool in the quick swap tool holder, and the workspace is clean

1.1.6. Once all these steps are done then you can turn on the machine by holding down the green power button on the control panel.



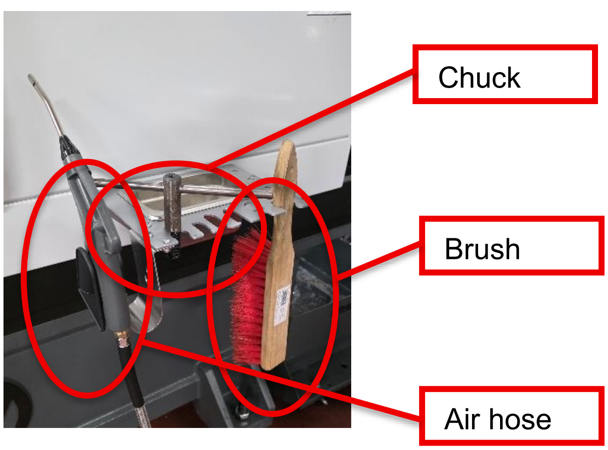
1.1.7.Once the machine is powered on you can verify that the **E-Stop button** is released by rotating it clockwise.

1.1.7.1. If the E-Stop button was pressed, then after release, press “reset” on the control panel.

1.1.7.2. verify that the doors are closed and that the control panel verified that the doors have been cycled (indicated with a green check mark on the panel screen)

1.1.7.3. Follow instructions on screen to home the tool.

1.6.Tools being used

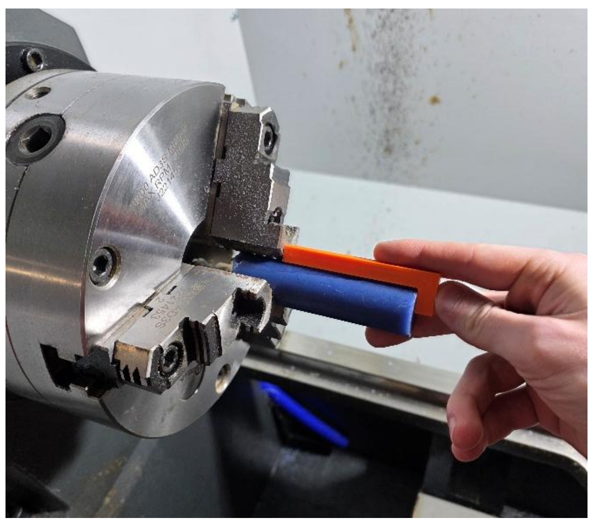


1.7.Loading the part:

1.7.1.Use the chuck key to open the chuck jaws to insert your part

1.7.2. Insert the desired stock material in the chuck

1.7.3. While supporting the stock material in the chuck, position the orange reference tool as shown. Slide the stock forward until its flat end contacts the reference tool. Hold the stock in position and use the chuck key to securely tighten the chuck jaws.

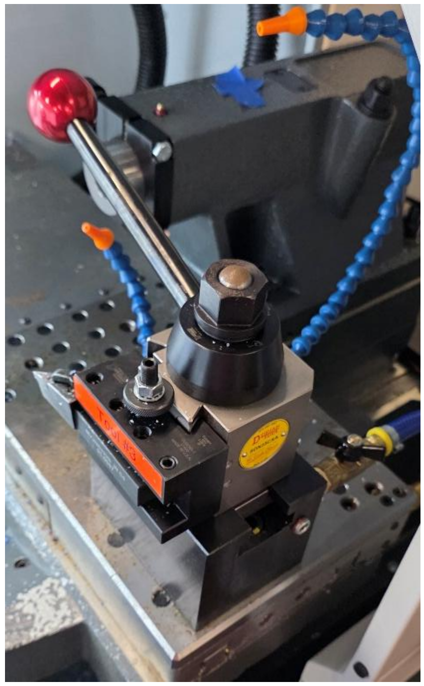


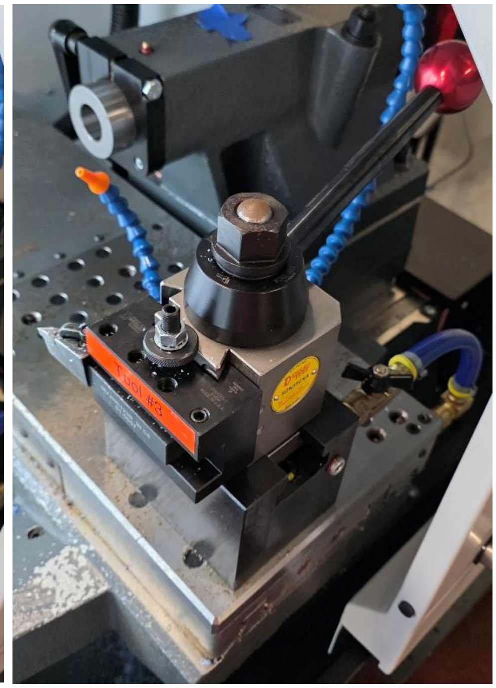
1.7.3.1. Safety Note: Make sure that your stock material isn't protruding more than 3 times its diameter from the chuck. If it is then start using the tailstock instructions to assist holding the stock material.

1.7.4. Make sure that you remove the chuck key from the chuck before proceeding.

1.8. Installing the tool:

1.8.1. Grab the tool #3 on the cart and insert it in the tool holder as shown.





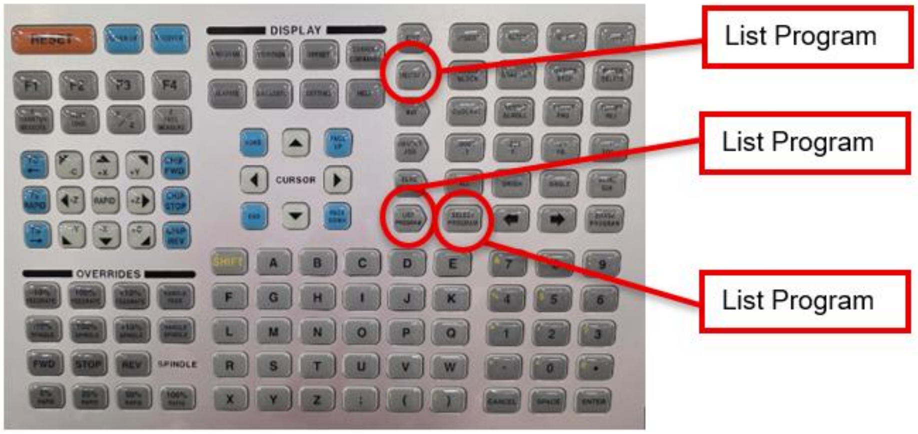
1.9.loading program:

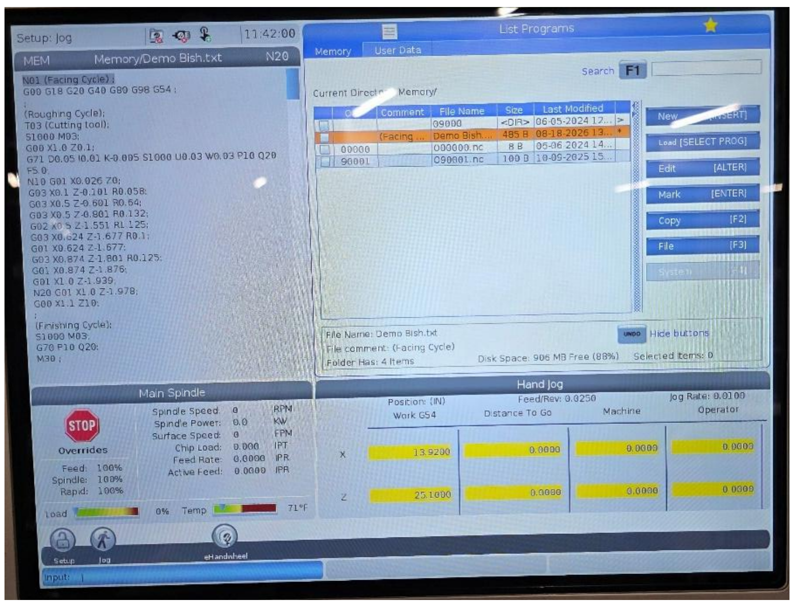
1.9.1.On the control panel select “List Program”

1.9.2.Use the cursor arrow keys to the

1.9.3.Once the desired code is highlighted then you press “select program”

1.9.4.Then select “Memory” to review your code and prepare it for operation





1.10. Final checks:

1.10.1. Once the code is loaded and "Memory" is selected you can use the arrow keys to scroll through the code to notice what tools and reference callouts are being made in the code.

1.11. Running the Program:

1.11.1. Once you verified then you are ready to run the program push "F3" and then push the green "Cycle Start" button on the control panel.

1.11.2. Recommended for running the program for the first, to slow the feed rate to 50% or slower depending on the machine task.

1.11.2.1. Safety Note: make sure you know where the E-stop button is located and be able to press it for any unexpected movements or collisions.

1.11.3. Once this process is complete you should have this

1.12. Cleaning work area:

1.12.1. Once the Program is done you can open the doors to observe your machined part.

1.12.2. Use gloves when handling any hot chips, and machined parts. Use a brush and air hose to push any chips off the tool bed.

1.12.3. Use the chuck key to remove the leftover stock material left in the chuck.

1.12.4. Make sure to put the brush, air hose, and chuck key back into their proper locations.

1.13. Powering off the machine

1.13.1. When powering off the CNC machine it is recommended that you zero the machine first to reduce the startup time for the next operator.

1.13.1.1. To zero the machine on the control panel you press “power up” followed by “A” this will automatically zero the machine

1.13.2. It is also recommended that you push in the E-stop as a safety precaution right before or after the machine is powered off

1.13.3. To power off the machine you have to hold down the red “power off” button on the control panel until the display turns off

1.13.3.1. The book doesn’t mention anything about flipping the power switch or air valve to the machine. I recommend powering off the machine once the machine operations are done for the day.